

— PORTFOLIO PROJECT 19 ·

2026-2027

Revenue Cycle Control Chain & Patient Financial Closure Intelligence™

A student-developed healthcare operations case study converting a simplified revenue-cycle list into a closed-loop control system spanning registration, eligibility, prior authorization, documentation, coding inputs, charge capture, claim submission, claim status, adjudication, remittance, denial management, payment posting, patient communication, and verified closure.

Created by Kori Pickle, BSHA
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Educational, simulated, and no PHI.
No real payer portals, claims, EHRs,
employer systems, patient records,
or reimbursement outcomes were
used.

— HEALTHCARE

OPERATIONS

INTELLIGENCE ENGINE™

Registration
Readiness
Claim
Remittance
Verified Closure

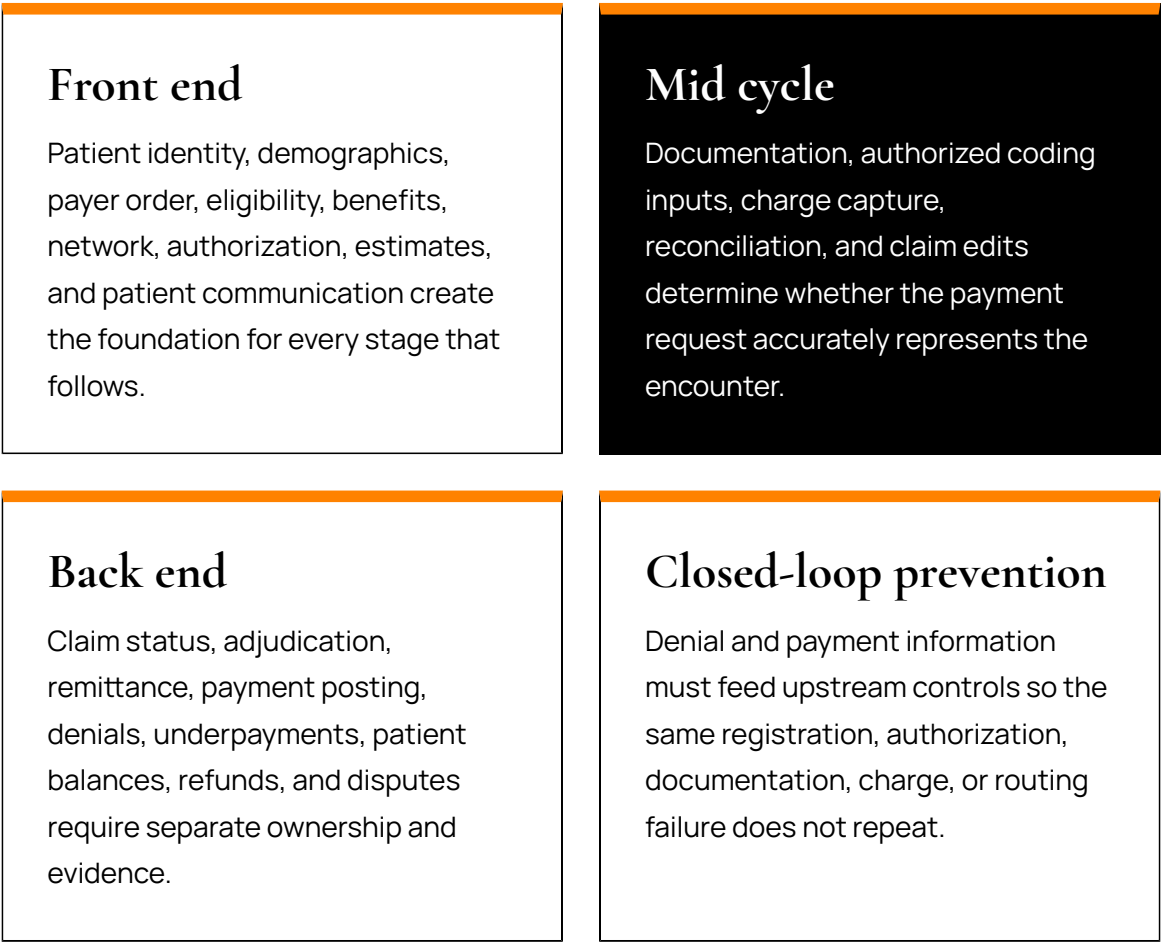
CORE QUESTION

Where
did the
workflow
first lose
control?

Kori Pickle · BSHA
Candidate

— EXECUTIVE BREAKDOWN

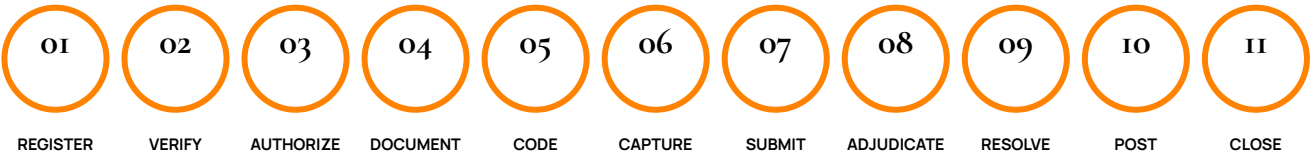
Revenue cycle management is a connected control chain, not one department.



A stage is complete only when its output is usable by the next owner and its unresolved risk remains visible.

ELEVEN-STAGE ARCHITECTURE

Translate each stage into an operational decision.



Stage	High-risk question	False-ready signal
Registration	Are patient, subscriber, payer order, service, provider, facility, date, and contact details aligned?	The account exists.
Eligibility and authorization	Do exact-plan, network, benefit, referral, authorization, and estimate controls match the final service?	Insurance is active.
Documentation and charge	Do orders, notes, services, units, supplies, authorization scope, and charge data reconcile?	The encounter has codes.
Claim and adjudication	Did the correct claim reach the correct payer, pass edits, and receive an owned status?	The claim was sent.
Posting and closure	Were payer and patient responsibilities validated, explained, resolved, and fed back upstream?	Payment posted.

SIMULATED CASE

One outpatient service. Six disconnected control failures.

A simulated patient is scheduled for advanced outpatient imaging. The organization uses separate scheduling, eligibility, authorization, clinical documentation, billing, and payment-posting workflows. Each team completes its local task, but no shared control confirms that the same patient, plan, service, location, date, units, and financial assumptions remain aligned.

Registration	The new plan is added, but the old payer remains primary.
Authorization	Approval was obtained for the original facility before the appointment moved.
Charge capture	

Claim outcome	The wrong payer receives the claim first; the corrected claim is denied for authorization-location mismatch.
Payment posting	The denial amount moves to patient responsibility before final internal review.
First loss of control	The coverage and location changes did not reset dependent authorization, network, estimate, routing, and posting statuses.

— CLOSED-LOOP REDESIGN

Eight controls from identity to verified closure.

1. Identify and validate

Confirm the patient, encounter, exact payer, payer order, service, provider, facility, date, communication needs, eligibility, benefits, and network.

2. Authorize and reconcile

Match approval evidence to the final service, provider, facility, dates, units, documentation, orders, charges, and quantities.

3. Submit and monitor

Release the correct claim to the correct payer, preserve submission evidence, monitor acknowledgments, claim status, aging, information requests, and timely-filing risk.

4. Resolve and close

Correct rejections, classify denials, review payment variances, protect patient balances, reconcile remittance and posting, communicate clearly, and update upstream controls.

The cycle closes with verified responsibility, accurate communication, and prevention feedback - not merely with a posted transaction.

CONTROL MATRIX

Revenue Cycle Control Chain Matrix™

Control	Required evidence	Status design	Reset or escalation trigger
Patient and encounter identity	Patient, subscriber, encounter, payer, provider, facility, service, date, contact needs	Confirmed / conflict / unresolved	Demographic, coverage, service, date, or location change
Payer and service readiness	Exact plan, payer order, eligibility, benefits, network, authorization, estimate assumptions	Ready / conditional / hold / changed	Dependency changes or evidence expires
Documentation and charge alignment	Orders, notes, logs, supplies, units, dates, providers, approved coding inputs	Reconciled / variance / pending	Missing, duplicate, conflicting, or unsupported item
Claim release and status	Correct payer, claim stream, edits, attachments, timely-filing date, acknowledgment, status	Submitted / rejected / pending / corrected	Edit failure, payer change, information request, aging threshold
Remittance and posting	835 or payer explanation, adjustments, payer responsibility, patient responsibility, variance review	Posted / variance / hold / review	Balance mismatch, underpayment, credit, unresolved denial
Patient financial closure	Validated balance, communication, payment	Explained / pending /	Unresolved payer issue, complaint,

Control	Required evidence	Status design	Reset or escalation trigger
	plan, assistance, refund, dispute, closure evidence	disputed / resolved	confusing bill, overpayment
Prevention feedback	Root cause, first loss of control, corrective action, owner, regression measure	Open / implemented / monitored / sustained	Repeated denial or workflow/system change

— SYNTHETIC UAT AND KPIS

Test the handoffs before a patient or claim absorbs the defect.

Coverage change

Dependent authorization, network, estimate, payer-order, and routing statuses reopen while prior evidence remains visible.

Facility change

Location-dependent network and authorization controls reset automatically.

Duplicate charge

Quantity reconciliation blocks claim-ready status until the variance is resolved.

Claim rejection

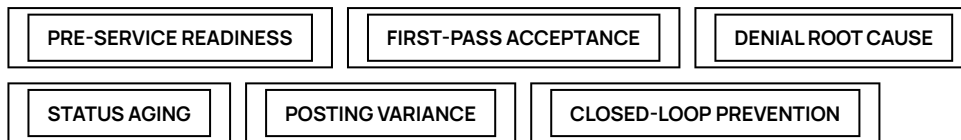
An owner, correction deadline, timely-filing risk, and resubmission evidence are required.

Denial to patient balance

Unresolved payer responsibility remains on internal hold until review is complete.

Information request

The request enters an exception queue with source, due date, approved document route, owner, and closure proof.



CAREER VALUE

Recruiter-visible proof for 2026-2027 roles.

Patient Access Identity, exact-plan verification, payer order, authorization, service readiness, and patient financial communication.	Revenue-Cycle Support Claim routing, status follow-up, denial root cause, remittance awareness, posting safeguards, and closure.	Implementation Support Business rules, dependency resets, exception queues, synthetic UAT, KPI design, and workflow adoption.
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I separate technology and departmental task completion from cross-stage evidence, ownership, dependencies, patient impact, and verified closure.

Official resources: [CMS Administrative Simplification Transactions](#); [CMS Claim Status](#); [CMS Payment and Remittance Advice](#); [CMS Interoperability and Prior Authorization Final Rule](#).
